Rick's wishlist

- Retire AI Assist (Data artifact v2 - Strikers + Mavericks)
- Design inveniam.io / hedgehog / swarm integration
- Data validation, consistency checking etc. (Strikers? - create pattern - continuation of Joe's work))
  - Event sourcing
    - Document the schema in confluence
    - internal consistency checker to run nightly
      - Purely internal
      - vs. Read db – including a
      - rebuild read db option
- Mongodb (Strikers)
  - Document the schema in confluence
  - Internal consistency checker program to run nightly
    - Purely internal
    - Foreign key consistency vs. Metachain, postgres, vector db
    - vs. FDR – validate URLs, hashes, whatever other metadata
    - vs Weaviate
  - Server enforced data integrity rules, domain checks, not null, unique indices, etc.
  - Evaluate indexing
  - Consider migration/correction of old data artifacts with invalid templates
- Weaviate (Mavericks)
  - Document the schema in confluence
  - Internal consistency checker to run nightly
    - Purely internal
    - Foreign key consistency vs. Metachain, postgres, mongodb
    - vs. FDR – validate URLs, hashes, whatever other metadata
- Metachain (Gadget)
  - Document the schema in confluence
  - internal consistency checker to run nightly

        - Purely internal
        - Foreign key consistency vs the others
        - vs. FDR
        - Vs. Chains
- Hub (Gadget)
    - Check every connection every night!
    - From HUB to blockchain
        - Validate that it works
        - Check balances
        - Report sent to devops and to hub admins for accounts with low balance
    - From inveniam.io to HUB
        - That the connection works
        - Reports sent in advance of credential expiration etc.
- Extend anchoring / Logging (Gadget)
    - Anchor more process steps e.g. extraction, tagging etc.
    - Anchor workflow actions
    - Log permissions violations, periodically anchor the logs.
    - Have agents anchor their work
        - What they did
        - What data they based their work on
        - Anchor of the output
    - Have aiChat log and anchor
    - Tie anchors together in sequence *(RGB is building a prototype for this)*
        - File uploaded
        - File tagged
        - File extracted
        - Data artifact created
        - Agent used the data
        - Agent decided something
        - Agent marked a workflow step

        - Agent created data
        - Human or agent used the data to create another document or artifact
        - That artifact can point back to its ancestor anchors
    - Extend anchoring of various logs
- Document Extraction Factory (Strikers? or Mavericks?)
    - Template creation wizard – AI assisted
    - Visual & text based analysis setup wizards
    - Prompt creation/testing/improvement tool – with immediate feedback
        - Rag
        - LLM
- New agents to production (Cypher)
    - Agent Factory! Decide on agentic framework, build reusable tools to make the creation of additional agents much easier
    - Memory management!!!!!
        - Inspect memory
        - Compare memory
        - Find problem memories
        - Test changes to memory
        - Curate/approve episodic memories, changes to memories
    - **Go to GP website download documents for an LP**
    - Automation of inveniam workflows
    - Automation of client activities
    - User feedback – learning agents / human in the loop
    - Prototyped agents -> production kyc/aml, policy alignment, email ingestion
- Database Interface Layer
    - Vector – expand to include Mongo, Pinecone etc.
    - NoSql – expand to include DocumentDB, Postgres and others
    - Graph – include the first, with interface layer for others to come
- APIs & Reporting (Nexus)
    - Apply permissions to the current FDR reader API
        - Extend ability to deliver data artifacts along with provenance

    - Create permissioned external API for engagement with workflow
    - Create permissioned external API to push data to Inveniam
    - Create permissioned APIs to support proof of origin, process, state
    - Extend FDR API to get artifacts from mongodb
    - Mongoreport prototype -> production
    - Graphs, charts, other reporting from mongodb
- MCP (Nexus)
    - In front of our public APIs
    - **With permissions**
- Data oracle
    - Allow a smart contract to ask for data
    - Whitelist of requestors/what data they can request
    - If allowed,
        - query inveniam,
        - deliver the data, encrypted with the requestors
        - Include the blockchain-anchored provenance of the data
- Optionality (some of this is not for Q4-2025)
    - ESPs – any more we need to add? I would love to add Intralinks if we could.
    - LLMs – test a bunch of the other compass models, open-weight openai ones
    - Blockchains (via hub) -- what more do we need to add?
    - Vector databases – pinecone, mongo vector
    - NoSQL databases – postgres, elastic search?
    - Enterprise wallets – Emanon, Fireblocks, Kaleido
    - Cloud providers – add GCP?
    - 3rd party agents – create permissioned MCP apis, SDKs for langgraph
    - Authentication
        - Future: authorization in active directory or via SAML claims